

Martin (2023) Single-Instrument Futures Momentum

\$50k capital · EWMAC 6 speeds equal-weight · per-trade skew characterisation

As of
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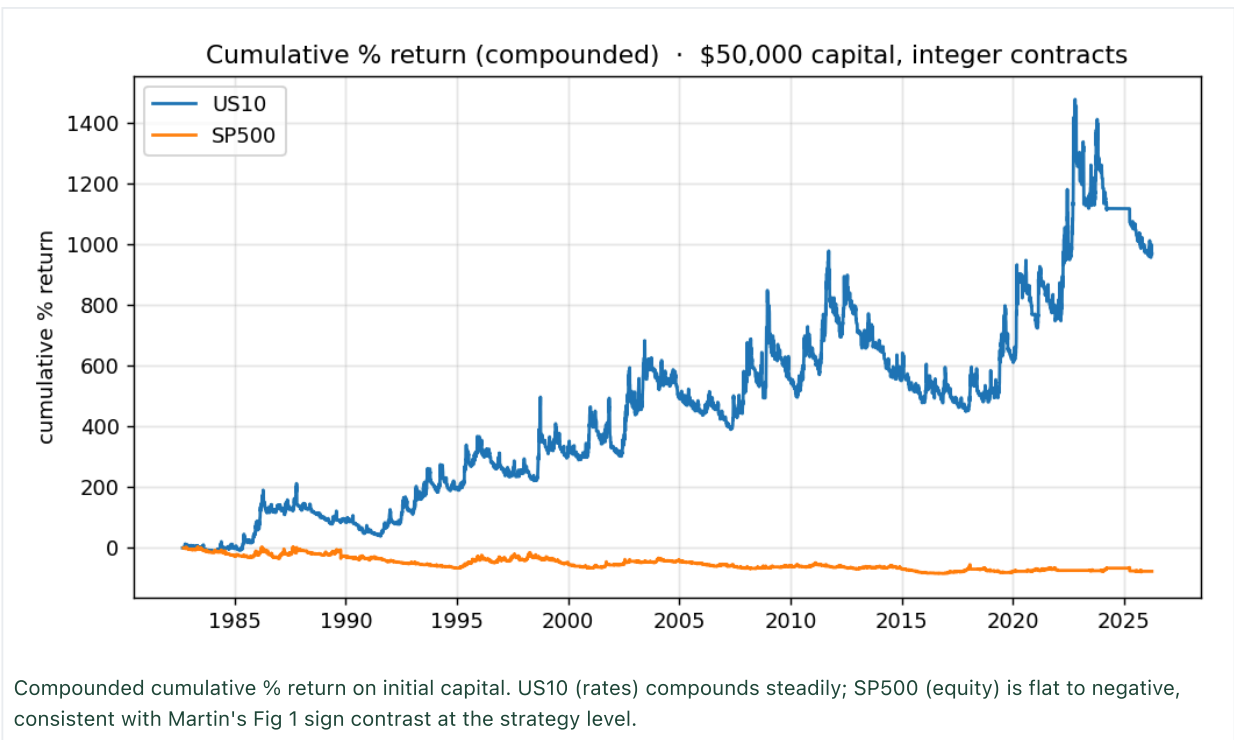
Headline Verification

- PASS** US10 market-return skew positive at M=40-60d [Martin Fig 1: rates side] *mkt skew @ M=60 = +0.204 (peak +0.24 at M=40)*
- PASS** SP500 market-return skew negative across horizons [Martin Fig 1: equity side] *mkt skew @ M=60 = -0.037 (negative for all M=1..40,100..250)*
- PASS** Per-trade skew >> daily skew on US10 [Martin §2.3: pure trend injects positive skew] *skew_per_trade = +4.38 vs daily +0.18*
- PASS** Low win rate + strong positive trade skew on US10 [Martin §3: long-option signature] *win% = 23.4 · skew_per_trade = +4.38*
- PASS** US10 gross Sharpe in [0.2, 0.6] [refutes 0.7-1.0 claim, matches Carver/Martin] *Sharpe_gross = 0.396*

Summary Metrics

instrument	ann_ret_g %	ann_vol %	Sharpe_g	Sharpe_n	skew_daily	maxDD % (geom)
US10	8.873	22.431	0.396	0.35	0.176	-55.347
SP500	-0.112	24.534	-0.005	-0.009	-1.988	-85.036

Equity Curves



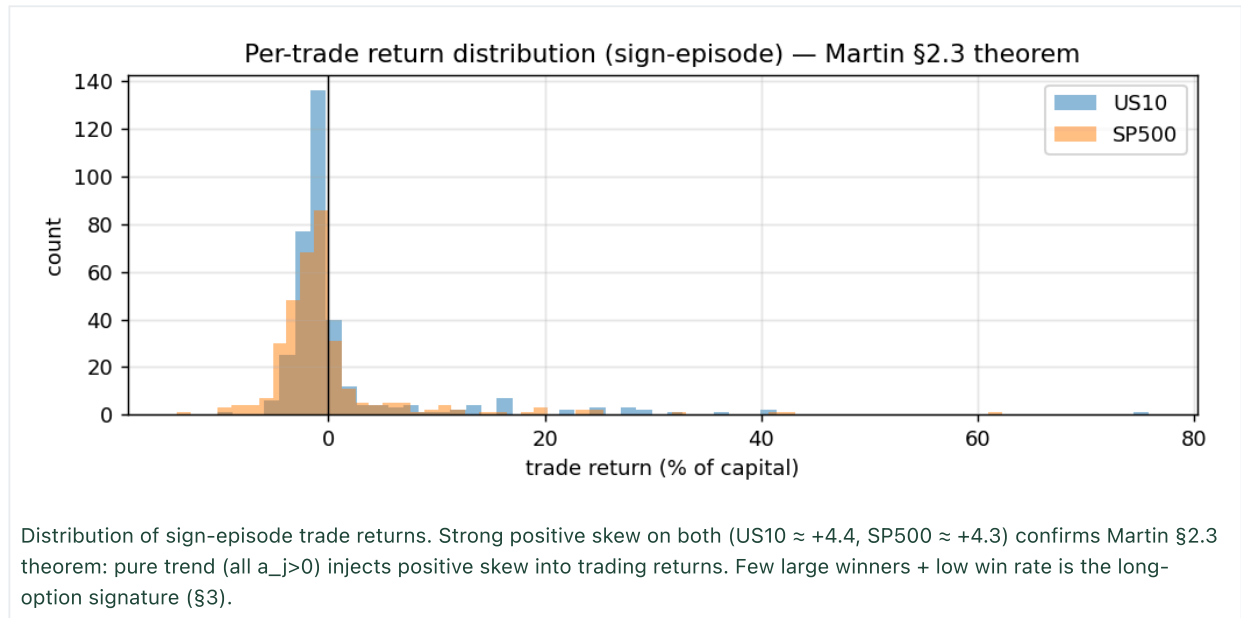
Literature: Richard J. Martin, "Design and analysis of momentum trading strategies" (2023). See [references/research/](#).

Per-Trade Analysis

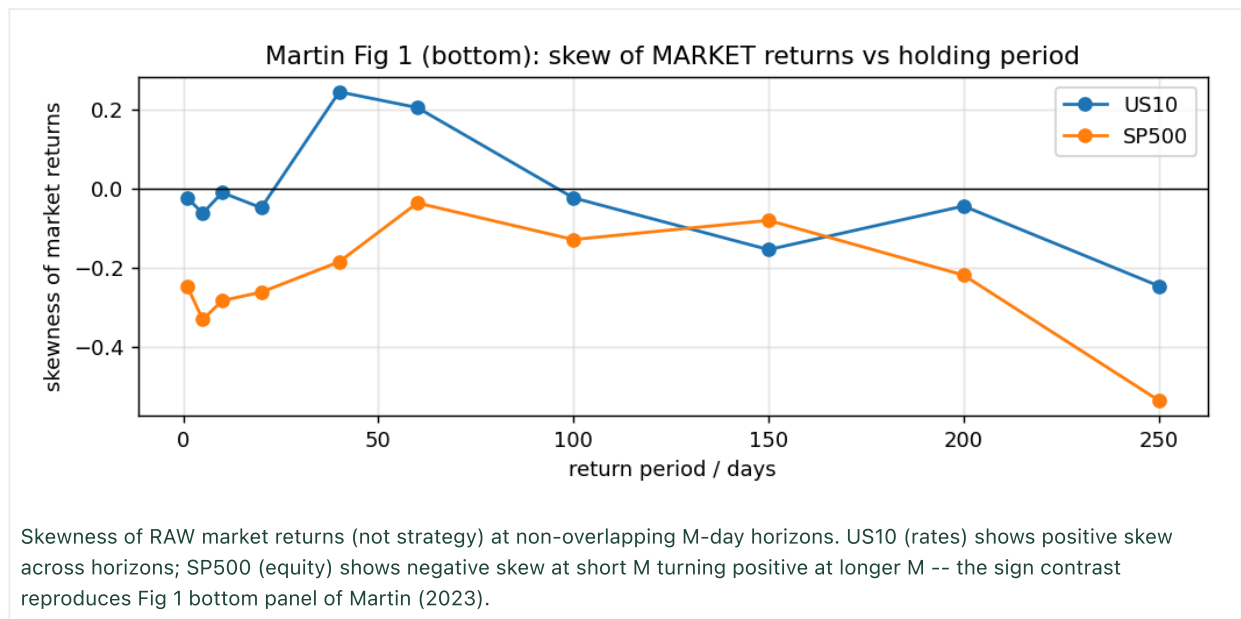
Sign-episode trade ledger · US10, SP500

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Per-Trade Distribution



Martin Fig 1 reproduction: market-return skew vs holding period



Trade Statistics

instrument	n_trades	win %	med days	med ret %	skew_trade	turn/yr
US10	342	23.4	12	-1.116	4.382	34.41
SP500	335	19.7	12	-1.306	4.327	33.47

Run Manifest

Engine	native pysystemtrade futures_chapter15 (do_not_reimplement_engine=true)
Instruments	US10, SP500

Trading rules	ewmac2_8, ewmac4_16, ewmac8_32, ewmac16_64, ewmac32_128, ewmac64_256
Forecast weights	equal (1/6 each), carry off, soft cap 20
Capital · Vol target · IDM	\$50,000 · 20.0% · 1.0
Trade definition	sign-episode: maximal run of constant non-zero sign(rounded buffered position)
git SHA	bb53383921bafd7d8769fc42d55195829a5e7e7f

